Kubernetes Task - 02

- Write a YAML file to create a Pod named firstpod with a nginx container. Verify the Pod creation using kubectl to get pods and check the logs of the container using kubectl logs firstpod.

```
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod

spec:
  containers:
    - name: nginx
      image: nginx
```

```
ubuntu@master:~/YAML$ kubectl apply -f firstpod.yml
pod/firstpod created
ubuntu@master:~/YAML$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
firstpod    1/1     Running   0          4s
```

- Modify the YAML file to include environment variables myname: Sabair and City: Hyderabad. Deploy the Pod and use kubectl exec <pod_name> -- env to check if the environment variables are set properly.

```
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod

spec:
  containers:
    - name: nginx
      image: nginx
      env:
        - name: myname
          value: "Sabair"
        - name: City
          value: "Hyderabad"
```

```
ubuntu@master:~/YAML$ kubectl apply -f firstpod.yml
pod/firstpod created
ubuntu@master:~/YAML$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
firstpod    1/1     Running   0          10s
ubuntu@master:~/YAML$ kubectl exec firstpod -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=firstpod
NGINX_VERSION=1.29.7
NJS_VERSION=0.9.6
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
City=Hyderabad
myname=Sabair
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
HOME=/root
```

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- Modify the YAML file to add args that instruct the container to sleep for 50 seconds. Deploy the Pod and use kubectl describe pod to verify the args are correctly passed to the container.

```
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod

spec:
  containers:
    - name: nginx
      image: nginx
      args: ["sleep", "50"]
ubuntu@master:~/YAML$ kubectl apply -f firstpod.yml
pod/firstpod created
ubuntu@master:~/YAML$ kubectl get pods
NAME       READY   STATUS    RESTARTS   AGE
firstpod   1/1     Running   0          6s
```

```
ubuntu@master:~/YAML$ kubectl describe pod firstpod
Name:             firstpod
Namespace:        default
Priority:         0
Service Account:  default
Node:             worker1/172.31.68.126
Start Time:       Sun, 05 Apr 2026 09:00:36 +0000
Labels:           <none>
Annotations:      cni.projectcalico.org/containerID: ddde32128badc278a4bad1
                  cni.projectcalico.org/podIP: 192.168.235.130/32
                  cni.projectcalico.org/podIPs: 192.168.235.130/32
Status:           Running
IP:               192.168.235.130
IPs:
  IP:  192.168.235.130
Containers:
  nginx:
    Container ID:  containerd://5f76e3e6f7989af80bac5c9a10e3aa6abcdb27c30a0
    Image:         nginx
    Image ID:      docker.io/library/nginx@sha256:7150b3a39203cb5bee612ff4a
    Port:          <none>
    Host Port:     <none>
    Args:
      sleep
      50
    State:          Running
```

- Create a YAML file to define a Pod with two nginx containers inside. Use kubectl exec to access both containers and verify that both containers can communicate through the same network (e.g., using telnet between them).

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```
ubuntu@master:~/YAML$ cat multipod.yml
apiVersion: v1
kind: Pod
metadata:
  name: multipod

spec:
  containers:
    - name: nginx1
      image: nginx

    - name: nginx2
      image: nginx
      command: ["/bin/sh", "-c"]
      args:
        - sed -i 's/listen       80;/listen       8080;/g' /etc/nginx/conf.d/default.conf && nginx -g 'daemon off;'
ubuntu@master:~/YAML$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE     IP                NODE      NOMINATED NODE   READINESS GATES
multipod   2/2     Running   0          5m36s   192.168.235.131   worker1   <none>           <none>
ubuntu@master:~/YAML$
```

```
ubuntu@master:~/YAML$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE     IP                NODE      NOMINATED NODE   READINESS GATES
multipod   2/2     Running   0          5m36s   192.168.235.131   worker1   <none>           <none>
ubuntu@master:~/YAML$ kubectl exec -it multipod -c nginx1 -- /bin/bash
root@multipod:/# hostname -I
192.168.235.131
root@multipod:/# exit
exit
ubuntu@master:~/YAML$ kubectl exec -it multipod -c nginx2 -- /bin/bash
root@multipod:/# hostname -I
192.168.235.131
root@multipod:/#
```

```
ubuntu@master:~/YAML$ kubectl exec -it multipod -c nginx1 -- /bin/bash
root@multipod:/# curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br />
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br />
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

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```
ubuntu@master:~/YAML$ kubectl exec -it multipod -c nginx2 -- /bin/bash
root@multipod:/# curl localhost:80
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br />
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br />
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

- Modify the YAML to include an init container that sleeps for 30 seconds before the main containers start. Verify the init container's execution using kubectl describe pod and check the logs to confirm its completion.

```
ubuntu@master:~/YAML$ cat multipod.yml
apiVersion: v1
kind: Pod
metadata:
  name: multipod

spec:
  initContainers:
    - name: init-sleep
      image: busybox
      command: ["sleep", "30"]

  containers:
    - name: nginx1
      image: nginx

    - name: nginx2
      image: nginx
      command: ["/bin/sh", "-c"]
      args:
        - sed -i 's/listen       80;/listen       8080;/g' /etc/nginx/conf.d/default.conf && nginx -g 'daemon off;'
ubuntu@master:~/YAML$ kubectl apply -f multipod.yml
pod/multipod created
ubuntu@master:~/YAML$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE   IP               NODE      NOMINATED NODE   READINESS GATES
multipod   2/2     Running   0          38s   192.168.189.70   worker2   <none>           <none>
```

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```
ubuntu@master:~/YAML$ kubectl describe pod multipod
Name:             multipod
Namespace:        default
Priority:         0
Service Account:  default
Node:             worker2/172.31.79.52
Start Time:       Sun, 05 Apr 2026 09:31:04 +0000
Labels:           <none>
Annotations:      cni.projectcalico.org/containerID: fd213d5ce3a53174bf835dac1b6387a9b049b561dcbfabd58f7b4030f07b7083
                  cni.projectcalico.org/podIP: 192.168.189.70/32
                  cni.projectcalico.org/podIPs: 192.168.189.70/32
Status:           Running
IP:               192.168.189.70
IPs:
  IP:  192.168.189.70
Init Containers:
  init-sleep:
    Container ID:  containerd://823fa7f1d53400ad70c915c7b0bd993bfd629d631c0ae5c45d2a5cab32689894
    Image:         busybox
    Image ID:      docker.io/library/busybox@sha256:1487d0af5f52b4ba31c7e465126ee2123fe3f2305d638e7827681e7cf6c83d5e
    Port:          <none>
    Host Port:     <none>
    Command:
      sleep
      30
    State:          Terminated
      Reason:       Completed
      Exit Code:    0
```

- Use the kubectl run nginx --image=nginx --dry-run=client -o yaml command to generate a Pod YAML definition. Modify the generated YAML to suit specific requirements (e.g., labels or environment variables) and deploy it.

```
ubuntu@master:~/YAML$ kubectl run firstpod --image=nginx --dry-run=client -o yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: firstpod
  name: firstpod
spec:
  containers:
  - image: nginx
    name: firstpod
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
ubuntu@master:~/YAML$ vi firstpod.yml
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: firstpod
    app: nginx
  name: firstpod

spec:
  containers:
    - image: nginx
      name: firstpod
      env:
        - name: myname
          value: "sabair"
        - name: City
          value: "Hyderabad"
  restartPolicy: Always
ubuntu@master:~/YAML$ kubectl apply -f firstpod.yml
pod/firstpod created
ubuntu@master:~/YAML$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE   IP               NODE      NOMINATED NODE   READINESS GATES
firstpod   1/1     Running   0          9s    192.168.189.71   worker2   <none>           <none>
ubuntu@master:~/YAML$
```

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- Create a YAML file to define a Pod. First, deploy it using kubectl create -f <file_name>.yml and then modify the YAML (e.g., change the image version). Use kubectl apply to redeploy and verify the difference between both commands.

```
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
  labels:
    app: nginx

spec:
  containers:
    - name: nginx
      image: nginx
ubuntu@master:~/YAML$ kubectl create -f firstpod.yml
pod/firstpod created
ubuntu@master:~/YAML$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE   IP               NODE      NOMINATED
firstpod   1/1     Running   0          8s    192.168.189.72   worker2   <none>
ubuntu@master:~/YAML$
```

```
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
  labels:
    app: nginx
    env: test

spec:
  containers:
    - name: nginx
      image: nginx
```

```
ubuntu@master:~/YAML$ kubectl create -f firstpod.yml
Error from server (AlreadyExists): error when creating "firstpod.yml": pods "firstpod" already exists
ubuntu@master:~/YAML$ kubectl apply -f firstpod.yml
Warning: resource pods/firstpod is missing the kubectl.kubernetes.io/last-applied-configuration annotation wh
ation will be patched automatically.
pod/firstpod configured
ubuntu@master:~/YAML$ kubectl get pod firstpod --show-labels
NAME       READY   STATUS    RESTARTS   AGE   LABELS
firstpod   1/1     Running   0          58s   app=nginx,env=test
```

- Use kubectl edit pod <pod_name> to modify the running Pod's environment variables or image. After making the changes, verify if they took effect by checking the container logs or environment variables using kubectl exec.

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```
ubuntu@master:~/YAML$ kubectl apply -f firstpod.yml
pod/firstpod created
ubuntu@master:~/YAML$ kubectl edit pod firstpod
pod/firstpod edited
ubuntu@master:~/YAML$ kubectl get pod firstpod --show-labels
NAME        READY   STATUS    RESTARTS   AGE   LABELS
firstpod    1/1     Running   0          50s   app=nginx,env=dev
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
  labels:
    app: nginx
    env: test

spec:
  containers:
    - name: nginx
      image: nginx
```

- Create a YAML file to expose your firstpod using a Service (ClusterIP). Ensure that your service is exposing the Pod on port 80 and verify it using kubectl get svc.

```
ubuntu@master:~/YAML$ cat firstpod.yml
apiVersion: v1
kind: Service
metadata:
  name: firstpod-service

spec:
  type: ClusterIP
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
ubuntu@master:~/YAML$ kubectl apply -f firstpod.yml
service/firstpod-service created
ubuntu@master:~/YAML$ kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
firstpod-service    ClusterIP   10.109.119.48   <none>        80/TCP    18s
kubernetes          ClusterIP   10.96.0.1       <none>        443/TCP   2d4h
```

- Add resource requests and limits to the containers in your YAML file. Specify CPU and memory requests/limits for both containers and deploy the Pod. Use kubectl describe pod to verify if the resource configurations are correctly applied.

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```
ubuntu@master:~/YAML$ cat multipod.yml
apiVersion: v1
kind: Pod
metadata:
  name: multipod

spec:
  containers:
    - name: nginx1
      image: nginx
      resources:
        requests:
          memory: "128Mi"
          cpu: "100m"
        limits:
          memory: "256Mi"
          cpu: "200m"

    - name: nginx2
      image: nginx
      command: ["/bin/sh", "-c"]
      args:
        - sed -i 's/listen       80;/listen       8080;/g' /etc/nginx/conf.d/default.conf && nginx -g 'daemon off;'
      resources:
        requests:
          memory: "128Mi"
          cpu: "100m"
        limits:
          memory: "256Mi"
          cpu: "200m"
ubuntu@master:~/YAML$ kubectl apply -f multipod.yml
pod/multipod created
ubuntu@master:~/YAML$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE   IP                NODE      NOMINATED NODE   READINESS GATES
multipod   2/2     Running   0          11s   192.168.235.132   worker1   <none>           <none>
```

```
    Restart Count:  0
    Limits:
      cpu:     200m
      memory:  256Mi
    Requests:
      cpu:        100m
      memory:     128Mi
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-jq8z6 (ro)
  nginx2:
    Container ID:  containerd://99a7dde46be2ac9f6c08b64f48fb229fc1b4228e28723713d5ae50bdb2cc605e
    Image:         nginx
    Image ID:      docker.io/library/nginx@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
    Port:          <none>
    Host Port:     <none>
    Command:
      /bin/sh
      -c
    Args:
      sed -i 's/listen       80;/listen       8080;/g' /etc/nginx/conf.d/default.conf && nginx -g 'daemon off;'
    State:          Running
      Started:      Sun, 05 Apr 2026 10:00:36 +0000
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     200m
      memory:  256Mi
    Requests:
      cpu:        100m
      memory:     128Mi
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-jq8z6 (ro)
Conditions:
  Type                        Status
```